Annotation Doc | Foundational Reading Tool | Jan 2026

Hindi | Transcribing Speech

Background

The primary objective of this annotation exercise is to create a labeled corpus of student reading recordings. This data will serve two purposes:

1. **Model Training:** To train Automatic Speech Recognition (ASR) models to transcribe recordings of a child reading a reference text created based on **Early Grade Reading Assessment (EGRA)** framework. The transcript generated by the ASR will be compared with the reference text to assess the reading.

2. **Model Evaluation:** To create a ground truth transcript which can be used to evaluate the ASR models.

Annotation Protocol

Inputs Provided to the Annotator

The annotator will be provided with the following elements for each task:

1. **Student audio:** A recording of a student reading a **Letter, a Word, or a Paragraph**.

2. **Reference Text:** The Letter, Word or Paragraph the student was instructed to read.

3. **Reference audios** (if available) **:** One or more **gold-standard readings** of the corresponding reference text.

Core Principle: Transcribe the whole audio as heard while respecting accent variations.

1. The **whole speech in the student audio needs to be transcribed** irrespective of what the reference text is. Though the reference text is a letter, the student audio may

contain a longer speech. Similarly the student audio can contain words or sounds that are not in the reference text. However, the transcript must be made for the whole audio.

2. If pronunciation of a word or letter in the reference text matches either, i) the standard pronunciation of that word or letter, or ii) the pronunciation of it in one of the reference audios (if available) , the transcript for it must be the same as the letter or word in the reference text. Otherwise, the word or letter needs to be transcribed as heard (verbatim). The **reference audios are provided as a contextual guide to accent variations** that the annotator may not be familiar with. In some cases reference audio(s) may not be available. In such cases, correctness needs to be decided solely based on step i) above.

Steps in the annotation process

1. **Timestamping:** Indicate the start and end-times of each word or letter in the transcript using a consistent time format (e.g. HH:MM:SS:MS). Please use an appropriate tool to accurately mark timestamps. Any section of the audio that is separated by a pause must be considered as a separate entity and timestamped. Indicate the start and end times of each word in the transcript using a consistent time format (e.g., HH:MM:SS:MS). Each word should fit snugly.
   I. In case of letter and word readings, if the audio is NOT longer than the reading of the letter or word, there is no need to timestamp the letter or word.

2. **Transcription:** While transcribing, the annotator should refer to the target text and should transcribe whatever she can hear to the best of the annotator's ability. There are following possibilities while transcribing the speech in the audio.

   A. **Correct Reading of a word or letter in the reference text :** If the pronunciation of a word or letter in the reference text matches the standard pronunciation, OR if it matches any of the pronunciations of the word in the reference audios, it is considered as a "correct" reading. In such a case, the transcript must be the same as the word or letter in the reference text.
      I. **DO copy and paste the correct words from the reference text**. For words or letters in the reference text that are spoken correctly (matching either the standard pronunciation or a pronunciation in one of the reference audios)) , instead of typing in the transcription, copy-paste the correct form from the given reference text itself. This would reduce typing errors in the transcription. For example, ज्ञान vs ग्यान, the correct spelling of the word is "ज्ञान," which can be easily mistaken with "ग्यान." To prevent such errors during word annotation, it is recommended to adhere to the reference text when the word's pronunciation has been verified as accurate.
      II. Examples:
         - If the word in the reference text is शक्तिमान but the child says सक्तिमान and this matches the pronunciation of the word in one of the reference audios, the transcript must be शक्तिमान.

   C. **Incorrect Reading or Substitution:** If the pronunciation of a word is not correct as per logic A) above, it is an incorrect reading or miscue.

I. Examples:
   - If the word in the reference text is शक्तिमान and the child read it as जक्तिमान and it is not matching with pronunciation of the word in any of the reference audios, transcribe it as जक्तिमान.
   - If the reference text is the letter “च” and what you hear is “ज”, the transcript must be ज.
   - If the reference text is the letter “च” and the audio says “मेरा<pause>न<pause>नाम<pause>कि<pause>रण<pause>च “, then the full speech needs to be transcribed and each individual token which is separated by pause needs to be timestamped separately.

D. **Inserted word:** Sometimes, the student might say a word (either valid in the language or not) that is not in the reference text. In such cases, it needs to be transcribed as heard.

E. **Splitting words while reading :** If a word is split into parts (say, individual letters or syllables or any subword unit) with a proper pause while speaking, but is correctly said overall, each part should be separately transcribed with relevant timestamps. As a rule of thumb, if there is <u>more than 25ms pause</u> within a word, the transcript must be split there.

I. Let’s say the reference word is कमल, but the child has said each letter with pauses, the transcript should have three letters with relevant timestamps, e.g., <क > <म > <ल >

II. Say the reference word is अनुशासन, but the child broke it into two syllables, the transcript should read <अनु> and <शासन>.

F. **Repetitions and false starts:** Include disfluencies such as repetitions and false starts in the transcript. For instance, while reading the word “बच्चा”, if the child says “ब ब बच्चा”, the transcript must be the same as what is said, including the repetitions and false starts.

Labels

There are five special labels. Note that for all of these labels, there must be an opening and closing tag. Secondly, if the label is applied to the transcript make sure each word or word-part (when word is splitted while speaking) needs to be tagged.

A. **Silences:** if there are pauses or silences longer than the usual intra-word pauses (say pause longer than 2 seconds), label that region of the audio with a <SIL> tag.

Silences: if there are pauses or silences longer than the usual intra-word pauses (say pause

i) For example, if 00:00:12:00 to 00:00:16:12 has no speech, add <SIL></SIL> to the portion.

B. **Fillers :** Include fillers, usually vocalised as ‘आ’. Please transcribe what is heard. Additionally, add the label <FIL> to **each word** in the portion.

I. For instance, if you hear “आ अ बा बाल बालक”, the annotations must include the following words and tags

- <FIL>आ</FIL>
- <FIL>अ</FIL>
- बा
- बाल
- बालक

In this example, the “बा” can sound like a filler sound. However, since the word being read is “बालक”, it is clearly resulting from sub-lexical reading of the word. Hence, the <FIL> tag is not added for “बा”.

C. **Mumbling and unintelligible speech:** If a part of the audio is unintelligible, then there is no need to transcribe the part. Please use the label <MB></MB> for this part of the audio.

i) For example, from 00:00:12:00 to 00:00:16:12 if you hear some mumbling or unintelligible speech, add <MB></MB> to the portion.

D. **Noise**: When there is noise or other sounds that are not speech (inhaling, coughing, ambient noise),

I. In case only background noise can be heard without any clear signs of the target speech, then <NOISE> can be used alongside the timestamp to indicate that the audio patch only contains unwanted background noise.

i) Example: If 00:00:13:00 to 00:00:14:12 has noise but there is no speech add <NOISE></NOISE> to the portion.

II. In case what is spoken can be heard despite the noise. Transcribe what is heard and add <NOISE> tag to each word.

i) Example: If the word "बालक" is heard with background noise it must be labelled as <NOISE>बालक</NOISE>.

E. **Adult voice:** If any of the words are spoken by an adult (sometimes the teacher prompts the child), add an <ADULT> tag to the word.

i) Example: If the word "बालक" is spoken by an adult, it must be labelled as <ADULT>बालक</ADULT>.

Note: These special labels are not mutually exclusive. There can be instances where more than one of the above cases arises. In such cases, all relevant labels need to be added. For instance, the child might use fillers in the presence of background noise. In that case, both <FIL> and <NOISE> labels need to be added.

Annotation Examples

Example-1

We have given below an illustrative example of how the annotation can be done. Please pay attention to the last column.

Audio sample: 2024-11-11_13-31-42.mp4

Target passage:
मैं अपने दादाजी के साथ रहती हूँ उन्हें मैं थाथा कहती हूँ लेकिन जब उन पर बहुत प्यार आता है तो मैं उन्हें थाथू बुलाती हूँ

Annotation
(Note: The timestamps shown are for illustrative purposes and may not be entirely accurate)

Start	End	Annotation
[1.072]	[1.462]	['मैं']
[1.485]	[1.808]	['<NOISE> अपने <NOISE/>']
[1.828]	[2.824]	['<NOISE> दादाजी <NOISE/>']
[3.22]	[3.511]	['के']
[3.546]	[4.076]	['<NOISE> साथ <NOISE/>']
[4.107]	[4.657]	['<NOISE> रहती <NOISE/>']
[4.733]	[5.023]	['हूँ']
[5.529]	[6.23]	['उन्हें']
[6.31]	[6.721]	['मैं']
[7.146]	[7.923]	['थाथ]
[8.023]	[8.599]	['कहती']
[8.634]	[9.014]	['हूँ']
[9.39]	[10.021]	['लेकिन']
[10.041]	[10.507]	['जब']
[10.542]	[11.073]	['उन']
[11.178]	[11.518]	['पर']
[11.588]	[12.144]	['बहुत']
[12.3]	[12.765]	['प्यार']
[12.785]	[13.371]	['आता']
[13.386]	[13.622]	['है']
[13.747]	[14.178]	['<NOISE> तो <NOISE/>']

[14.328]	[14.503]	['<NOISE> मैं <NOISE/>']
[14.513]	[15.224]	['<NOISE> उन्हें <NOISE/>']
[15.395]	[16.196]	['<NOISE> थाथू <NOISE/>']
[16.667]	[17.583]	['<NOISE> बुलाती <NOISE/>']
[17.608]	[17.848]	['हूँ']

Example-2

Audio sample: 2024-11-11_14-11-21.mp4

Target passage:
नीतू के घर में गाय है उसका रंग सफेद है गाय हरी घास खाती है वह बहुत दूध देती है

Annotation
(Note: The timestamps shown are for illustrative purposes and may not be entirely accurate)

Start	End	Annotation
[0.628]	[1.507]	['<NOISE> नीतू <NOISE/>']
[1.584]	[1.939]	['<NOISE> के <NOISE/>']
[1.969]	[2.327]	['<NOISE> धर <NOISE/>']
[2.364]	[2.752]	['<NOISE>में<NOISE/>']
[2.789]	[3.306]	['<NOISE> गया <NOISE/>']
[3.333]	[3.651]	['<NOISE> है <NOISE/>']
[3.915]	[4.691]	['<NOISE> उसका<NOISE/>']
[4.702]	[5.234]	['<NOISE> रंग <NOISE/>']
[5.245]	[5.906]	['<NOISE> सफेद <NOISE/>']
[5.913]	[6.194]	['है']
[6.246]	[6.585]	['<NOISE> गय <NOISE/>']
[6.766]	[7.287]	['<NOISE> गय <NOISE/>']
[7.365]	[7.948]	['गा']
[8.059]	[8.89]	['<NOISE> गाईर <NOISE/>']
[9.104]	[9.699]	['<NOISE> हरी <NOISE/>']
[9.714]	[10.316]	['<NOISE> घास <NOISE/>']
[10.331]	[10.84]	['<NOISE> खाती <NOISE/>']
[10.848]	[11.258]	['<NOISE> है <NOISE/>']
[11.562]	[11.976]	['<NOISE> वह <NOISE/>']
[11.993]	[12.65]	['<NOISE> बहुत <NOISE/>']

Output Format

Json file with following format. Given below, is an example of the expected json output for an audio sample for reference


```
{
    "id": 13945,
    "file_name": "audio_student_2366399046006104195.wav",
    "annotations": [
        {
            "start": "0:00:00.519350",
            "end": "0:00:01.209974",
            "Transcription": [
                "शेर"
            ],
        },
        {
            "start": "0:00:01.232074",
            "end": "0:00:01.651974",
            "Transcription": [
                "जंगल"
            ],
          ]....
.
.
.
.
.
        },
```